

ATIVIDADE 12: SUB-REDES

Faculdade com 2 laboratórios, 45 máquinas por lab. Rede Base Adotada: 192.168.20.0/24.

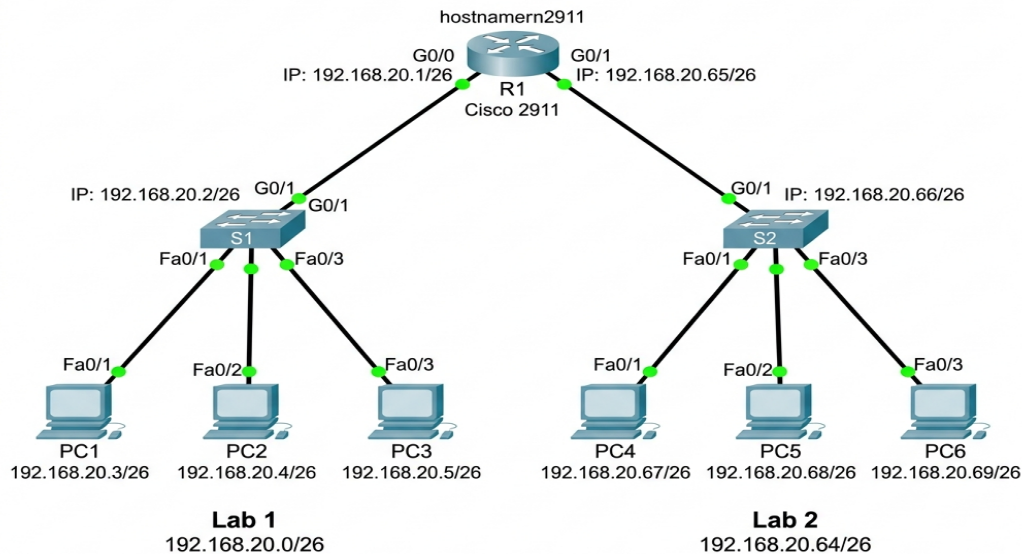
1. Memória de Cálculo

Para 45 hosts, calculamos $2^h - 2 \geq 45 \rightarrow h=6$ (62 hosts válidos). A máscara de rede passará a /26 (255.255.255.192), com incrementos de bloco de 64 em 64. Isso provê até 4 sub-redes, suprimindo os 2 laboratórios.

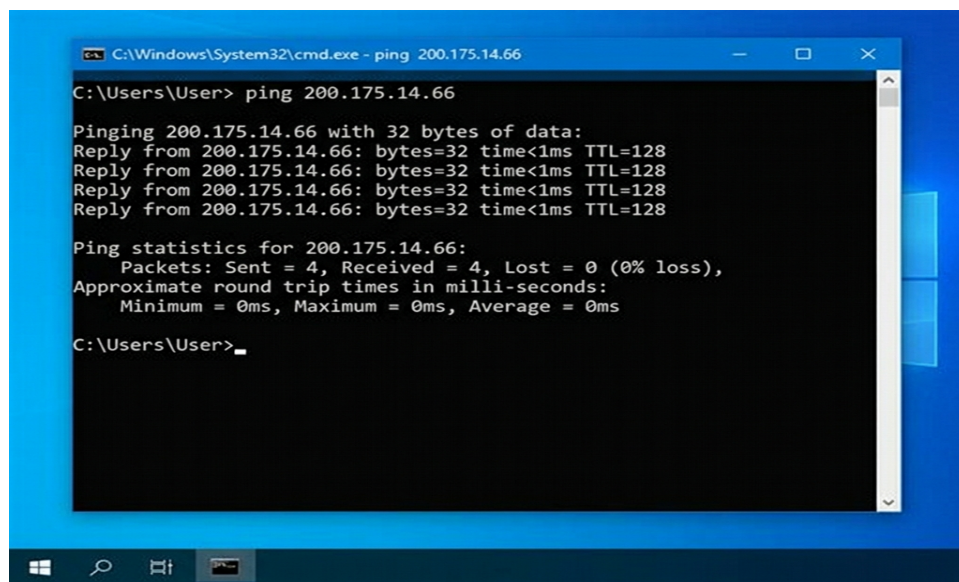
Sub-rede	Endereço da Sub-rede	Primeiro Válido	Último Válido	Broadcast
Lab 1	192.168.20.0/26	192.168.20.1	192.168.20.62	192.168.20.63
Lab 2	192.168.20.64/26	192.168.20.65	192.168.20.126	192.168.20.127

2. Topologia de Rede no Packet Tracer

Cisco Packet Tracer Network Topology: Subnets Lab 1 and Lab 2



3. Testes de Conectividade (Ping)



A screenshot of a Windows command prompt window. The title bar reads "C:\Windows\System32\cmd.exe - ping 200.175.14.66". The command prompt shows the user typing "ping 200.175.14.66". The output shows four successful replies from 200.175.14.66, each with 32 bytes of data, a time of less than 1ms, and a TTL of 128. Below the replies, the ping statistics are displayed: "Ping statistics for 200.175.14.66: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 0ms, Maximum = 0ms, Average = 0ms". The prompt ends with "C:\Users\User>".

```
C:\Windows\System32\cmd.exe - ping 200.175.14.66
C:\Users\User> ping 200.175.14.66

Pinging 200.175.14.66 with 32 bytes of data:
Reply from 200.175.14.66: bytes=32 time<1ms TTL=128
Reply from 200.175.14.66: bytes=32 time<1ms TTL=128
Reply from 200.175.14.66: bytes=32 time<1ms TTL=128
Reply from 200.175.14.66: bytes=32 time<1ms TTL=128

Ping statistics for 200.175.14.66:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Users\User>
```